

Probability as the Language of Uncertainty

Part 1: How do we represent information and uncertainty mathematically?

Probabilistic Machine Learning

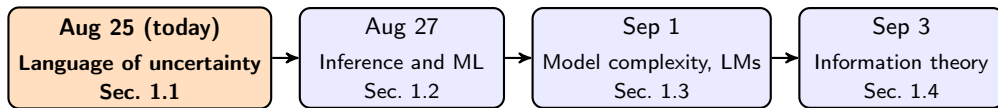
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Outline

- 1 Uncertainty, and what probability does about it
- 2 The language: what Chapter 0 gives us
- 3 Probabilistic reasoning: a worked example

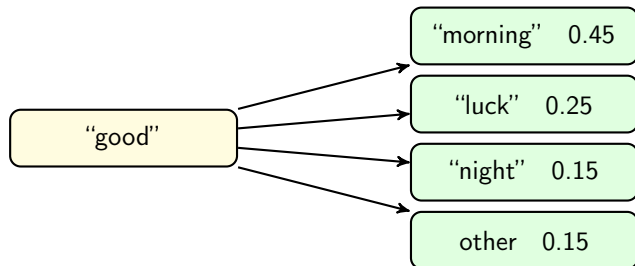
Roadmap of Part 1



Chapter 0 is the reference for the tools we use throughout (review test Sep 3).

How do we represent uncertainty?

When you type “good” into your phone, the keyboard suggests likely continuations.



(Numbers are illustrative.)

Distributions represent uncertainty

A distribution over the next word records both what the options are and how strongly each is expected, and what the phone does not know.

Three tasks probability serves

This course is concerned with three main problems, in a Probabilistic framework:

PROBABILISTIC REASONING

Given distributions for a set of variables, draw conclusions about unobserved variables from observed ones.

INFERENCE

Draw conclusions about the real world from evidence, for example whether a gene is linked to a disease.

SUPERVISED MACHINE LEARNING

Learn from (input, output) pairs and predict the output for new inputs, for example house prices.

- These definitions are not universal.
- Machine learning cares more about prediction accuracy than about representing reality.
- We will also discuss unsupervised machine learning and generative AI; this overview lecture is limited to supervised machine learning.

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Finite sample size

How height depends on the chosen factors is determined from data, and finite data cannot determine it fully: some combinations of the factors appear only a few times, or never.

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Sample space, events, axioms

Let's do an extremely brief review:

Definitions

Ω : the sample space, the set of all outcomes. $E \subseteq \Omega$: an event.

Axioms

$\Pr(E) \geq 0$ $\Pr(\Omega) = 1$ $\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2)$ if $E_1 \cap E_2 = \emptyset$

Two consequences

If $A \subseteq B$, then $\Pr(A) \leq \Pr(B)$. $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$.

What does $\Pr(E) = 1/3$ mean?

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Interpretation	Meaning	Example	Criticism
Classical	symmetry among outcomes	rolling a 3: $1/6$	circular
Frequentist	long-run frequency	$\Pr(H) = 1/3$: about 1000 heads in 3000 tosses	vague
Bayesian	degree of belief	a defendant is guilty	subjective

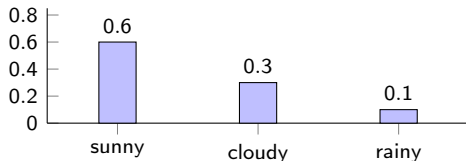
Axioms and interpretations

Mathematically, only the axioms matter, but interpretation shapes how problems are approached.

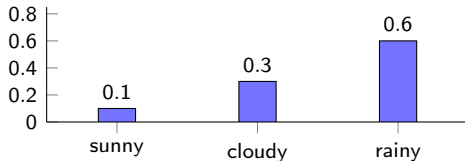
A distribution is a state of knowledge (and uncertainty)

The distribution of a random variable tells us what its possible values are and how likely they are. It represents both our state of knowledge and our uncertainty about the variable.

Tomorrow's weather, from the season's averages



After radar shows a storm front



The distribution is our current state of knowledge; new information changes it.

Joint distributions encode relationships

Two flips of a fair coin. $X = 1$ if the first flip is heads, $Y =$ total number of heads.

$p(x, y)$	$y = 0$	$y = 1$	$y = 2$	$p_X(x)$
$x = 0$	1/4	1/4	0	1/2
$x = 1$	0	1/4	1/4	1/2
$p_Y(y)$	1/4	1/2	1/4	

$$p_X(x) = \sum_y p(x, y)$$

The marginals sit in the margins of the table, which is where the name comes from.

Four rules we use today

- Conditional probability / product rule: $\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$, equivalently $p(x, y) = p(x)p(y|x)$; for three variables, $p(x, y, z) = p(x)p(y|x)p(z|x, y)$
- Total probability: $\Pr(B) = \sum_i \Pr(B|A_i) \Pr(A_i)$
- Bayes' rule: $p(x|y) = \frac{p(y|x)p(x)}{p(y)}$
- Independence: $A \perp\!\!\!\perp B$ iff $\Pr(AB) = \Pr(A) \Pr(B)$, equivalently $\Pr(A|B) = \Pr(A)$

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Alice, Bob, and the traffic

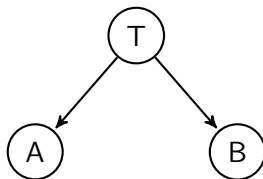
Two coworkers live in the same area; the business opens at 9:00. Are their arrival times related? How do we represent the relationship? Can we predict one being late from the other?

T_O : normal traffic, T_H : heavy traffic

A_O, A_L : Alice on time / late

B_O, B_L : Bob on time / late

Modeling assumption: given the traffic, Alice and Bob are independent.



$$\Pr(T_O) = 0.75, \quad \Pr(A_O|T_O) = 0.8$$

$$\Pr(B_O|T_O) = 0.6, \quad \Pr(A_O|T_H) = 0.6$$

$$\Pr(B_O|T_H) = 0.2$$

Five numbers determine everything

T	A	B	$p(t, a, b)$
T_O	A_O	B_O	0.36
T_O	A_O	B_L	0.24
T_O	A_L	B_O	0.09
T_O	A_L	B_L	0.06
T_H	A_O	B_O	0.03
T_H	A_O	B_L	0.12
T_H	A_L	B_O	0.02
T_H	A_L	B_L	0.08

$$p(t, a, b) = p(t) p(a|t) p(b|t)$$

A general joint over three binary variables needs 7 numbers. Why do 5 suffice here?

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A general joint over three binary variables needs 7 numbers. Why do 5 suffice here?

Assumptions reduce the number of parameters

The product rule gives $p(t, a, b) = p(t) p(a|t) p(b|t, a)$; the assumption $A \perp\!\!\!\perp B \mid T$ drops a from the last factor, leaving $1 + 2 + 2 = 5$ numbers instead of 7.

Causal reasoning: from causes to effects

Your turn: what is the probability Alice is on time? Bob?

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$$\Pr(A_O) = \Pr(T_O) \Pr(A_O|T_O) + \Pr(T_H) \Pr(A_O|T_H) = 0.75(0.8) + 0.25(0.6) = 0.75$$

$$\Pr(B_O) = \Pr(T_O) \Pr(B_O|T_O) + \Pr(T_H) \Pr(B_O|T_H) = 0.75(0.6) + 0.25(0.2) = 0.5$$

This is the **law of total probability**: sum over the ways the cause can occur.

Evidential reasoning: from effects to causes

Bayes' rule turns the numbers around:

$$\Pr(T_O|A_O) = \frac{\Pr(A_O|T_O) \Pr(T_O)}{\Pr(A_O)} = \frac{0.8 \times 0.75}{0.75} = 0.8$$

$$\Pr(T_O|B_O) = \frac{\Pr(B_O|T_O) \Pr(T_O)}{\Pr(B_O)} = \frac{0.6 \times 0.75}{0.5} = 0.9$$

Not all evidence is equally strong

Bob's punctuality is stronger evidence about the traffic than Alice's: his on-time rate swings from 0.6 to 0.2 with traffic, hers only from 0.8 to 0.6.

Are Alice and Bob independent?

$$\Pr(A_O B_O) = \Pr(T_O) \Pr(A_O | T_O) \Pr(B_O | T_O) + \Pr(T_H) \Pr(A_O | T_H) \Pr(B_O | T_H) = 0.39$$

$$\Pr(A_O | B_O) = \frac{0.39}{0.5} = 0.78 \neq \Pr(A_O) = 0.75$$

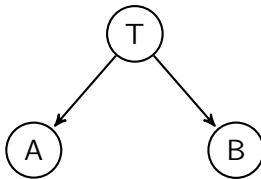
$$\Pr(B_O | A_O) = \frac{0.39}{0.75} = 0.52 \neq \Pr(B_O) = 0.5$$

$A_O \not\perp B_O$. A common cause makes them dependent.

...but conditionally independent given the traffic

By assumption,

$$\Pr(A_O B_O | T_O) = \Pr(A_O | T_O) \Pr(B_O | T_O), \quad \text{written } A_O \perp\!\!\!\perp B_O | T_O.$$



Conditional independence

The dependence between A and B runs entirely through T . Once T is known, one tells you nothing more about the other.

Your turn

(a) Does knowing Alice is late change Bob's distribution? Compare $\Pr(B_L)$ and $\Pr(B_L|A_L)$.

(b) The traffic is known to be heavy. Does knowing Alice is late still change Bob's distribution?

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(a) Does knowing Alice is late change Bob's distribution? Compare $\Pr(B_L)$ and $\Pr(B_L|A_L)$.

$\Pr(A_L) = 0.25$, $\Pr(A_L B_L) = 0.75(0.2)(0.4) + 0.25(0.4)(0.8) = 0.14$, so

$$\Pr(B_L|A_L) = \frac{0.14}{0.25} = 0.56, \quad \text{up from } \Pr(B_L) = 0.5.$$

(b) The traffic is known to be heavy. Does knowing Alice is late still change Bob's distribution?

$\Pr(B_L|A_L, T_H) = \Pr(B_L|T_H) = 0.8$: no; once traffic is known, Alice's lateness adds nothing.

What is the source of uncertainty here?

The distribution is fully known, so finite sample size is not the issue. The source is **noise**: Bob's car, whether he drops off his kids. Modeling more factors would reduce it.

Task	Source of uncertainty
Probabilistic reasoning	noise only
Inference	finite sample size
Supervised machine learning	both

The next lecture works out inference and machine learning.

What probability did for us

- Represented knowledge: five numbers specified the entire model.
- Encoded an assumption: conditional independence given the traffic.
- Gave rules for updating with evidence: Bayes' rule.

From probability to learning

When the distributions are known, probability alone answers every question. When they are not, we must learn them from data.

- Probability represents knowledge and uncertainty through distributions.
- The axioms are fixed; interpretation (classical, frequentist, Bayesian) shapes how we approach problems.
- Joint distributions and conditional independence let a few numbers determine an entire system.
- Bayes' rule turns causal knowledge into evidential reasoning.

Next: inference and machine learning, Sec. 1.2.